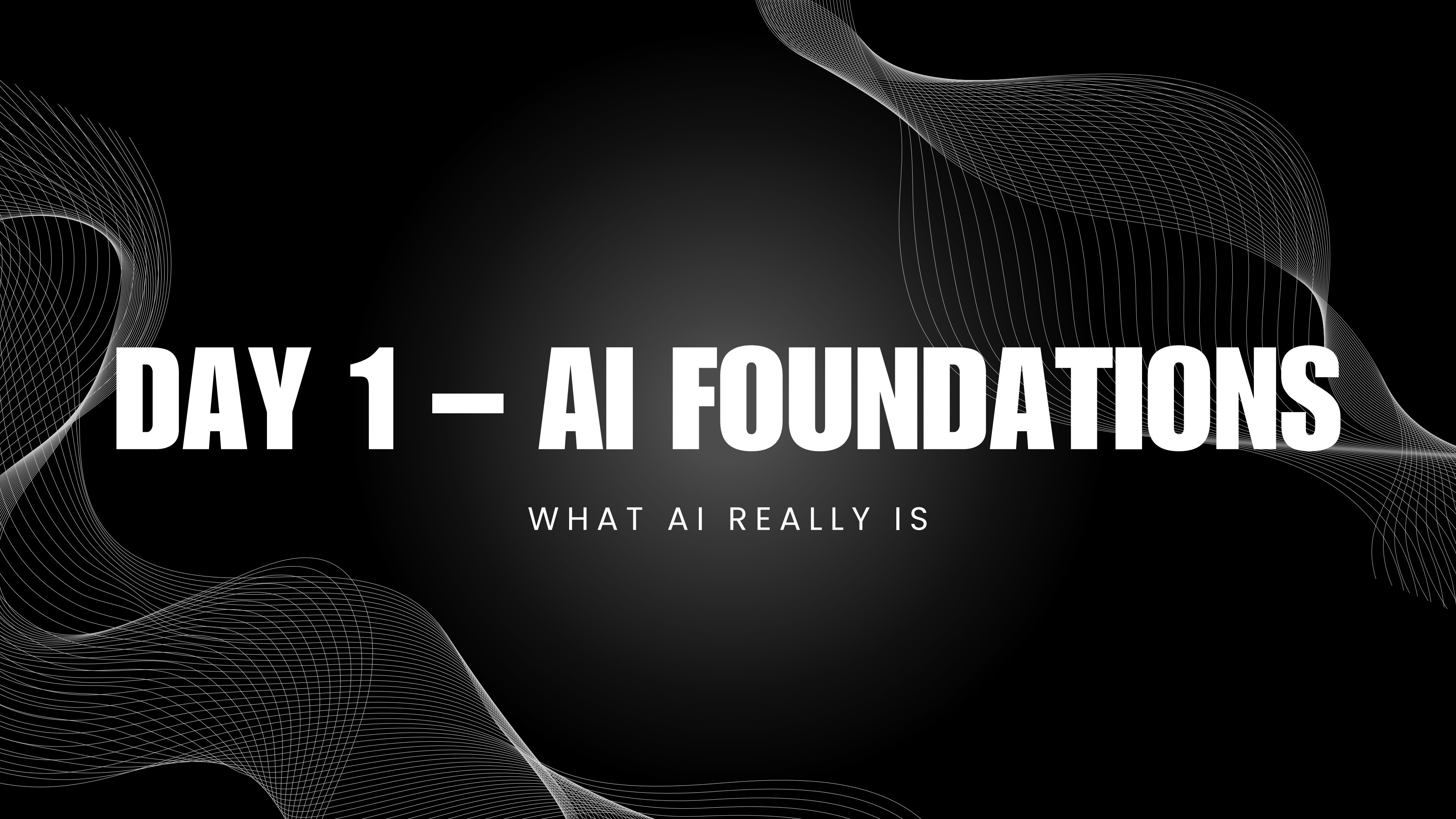




DAY 1 — AI FOUNDATIONS

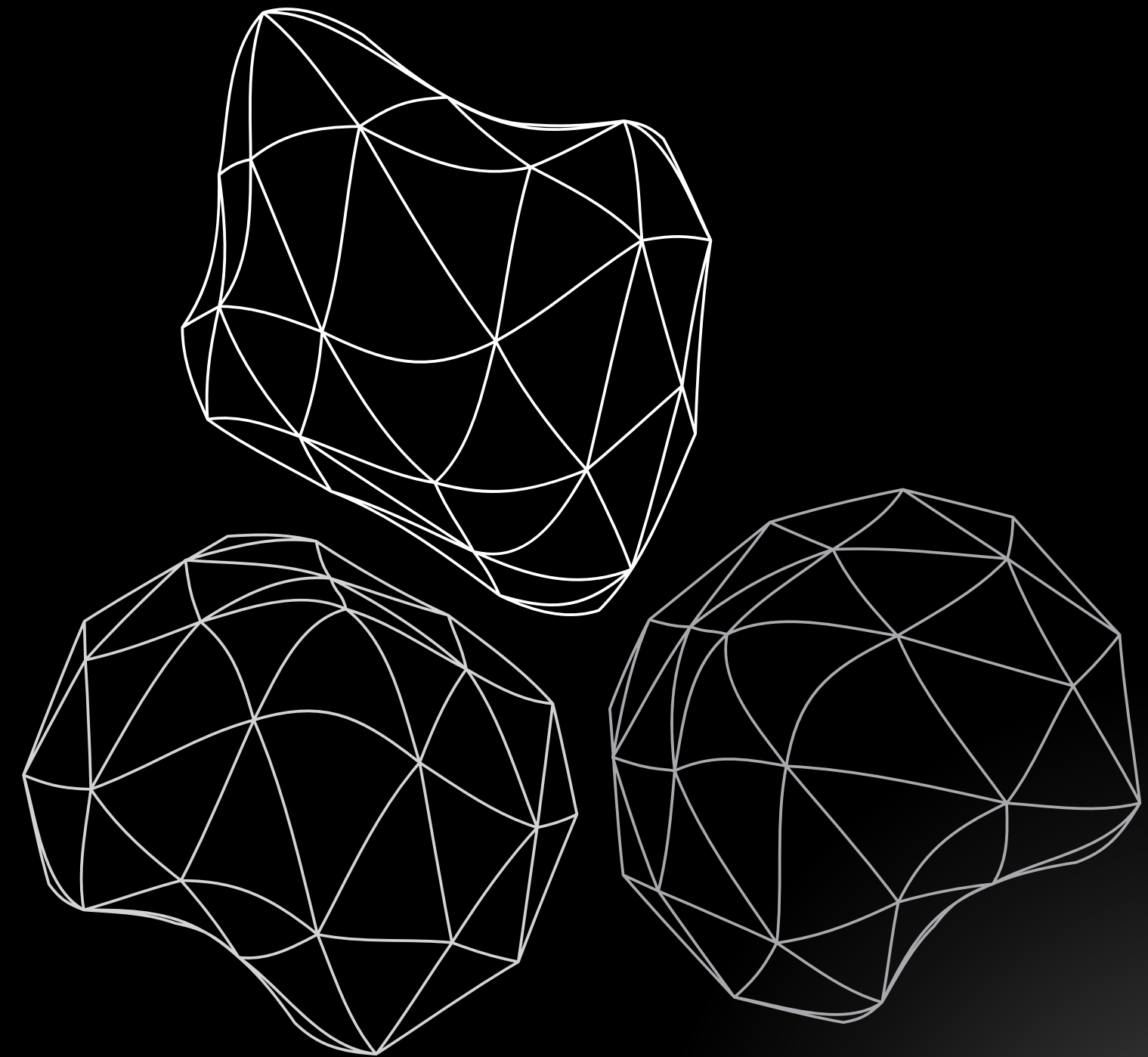
WHAT AI REALLY IS

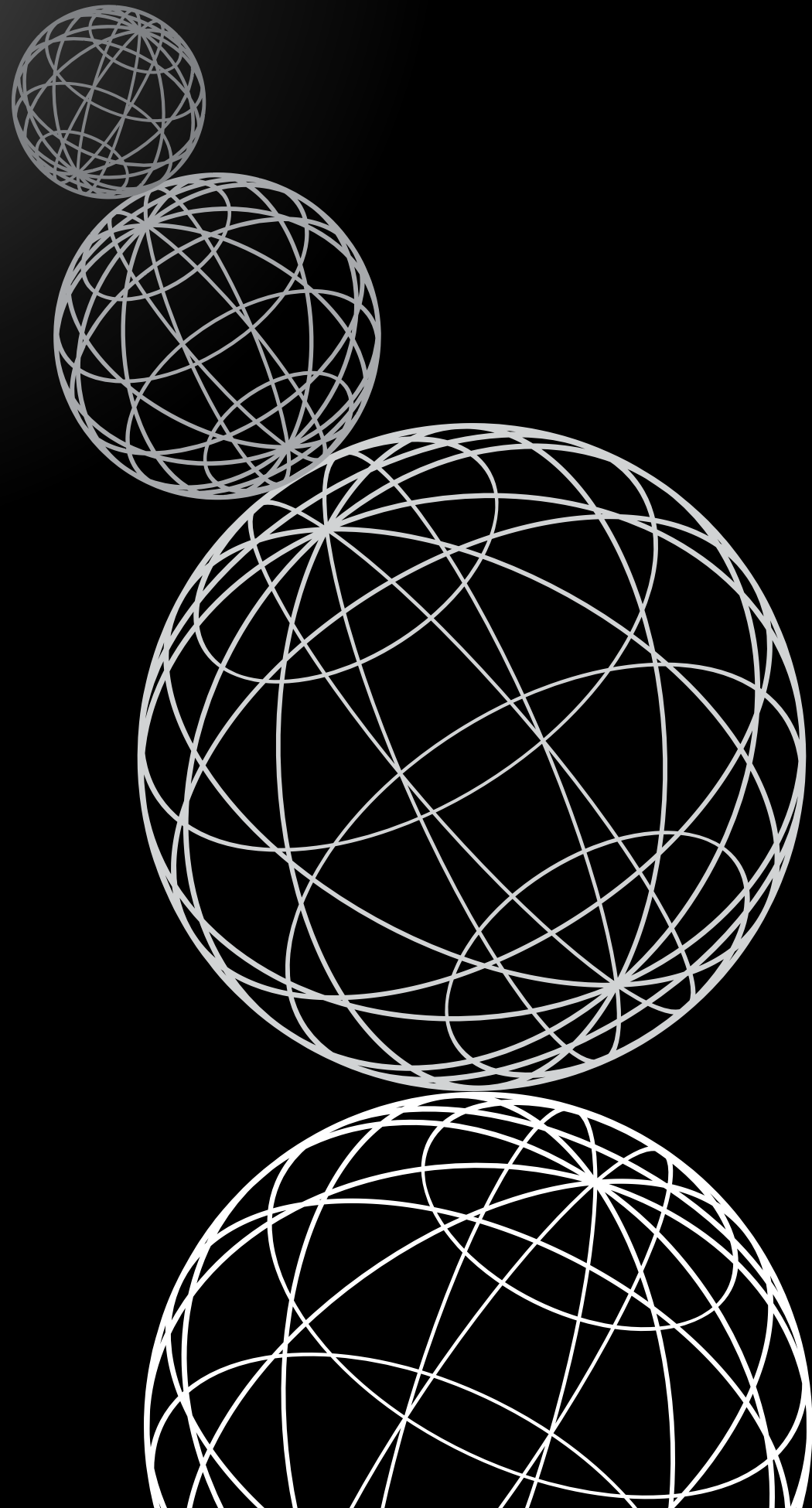
AI FOUNDATIONS

WHAT AI REALLY IS (AND WHAT IT IS NOT)

Purpose of today:

- ✦ Reset mental models
- ✦ Remove hype
- ✦ Build correct AI thinking from day one





WHY ARE WE EVEN DOING THIS?

CODE SKILLS

You already know how to call AI / LLM APIs and understand basic NLP terminology, enabling you to effectively integrate AI into your products and services.

UNDERSTANDING GAPS

Yet, AI requirements feel unclear, decisions often appear guessy, and interviews frequently expose gaps in understanding, highlighting the need for clarity beyond just coding skills.

80%

Engineers' Struggles with AI

Most engineers face challenges in identifying AI problems, selecting appropriate types, and justifying their decisions, highlighting the need for improved understanding and communication in AI applications.



What This Program Is NOT

Our approach to AI education

We are NOT

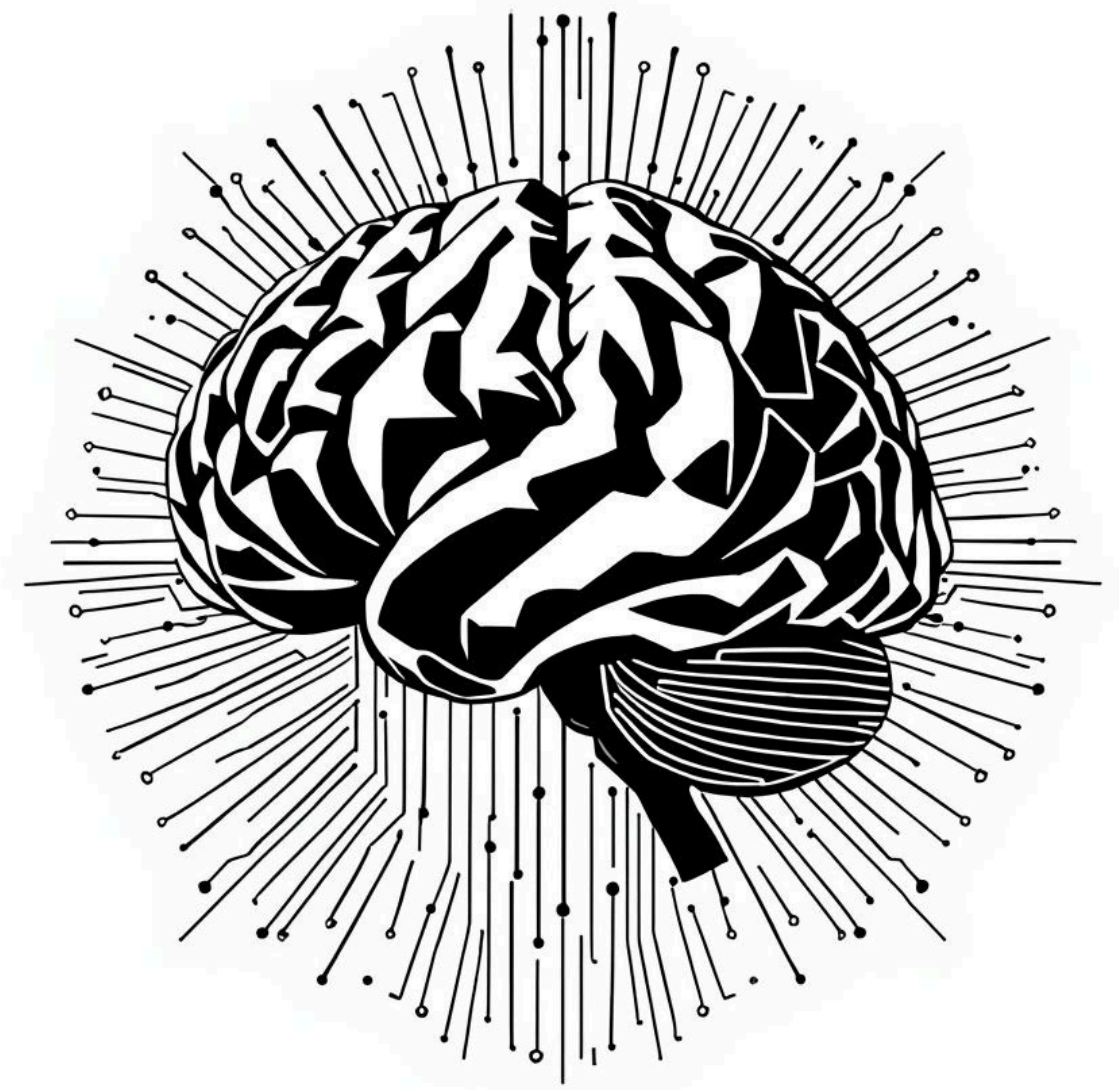
This program does not aim to turn you into mathematicians, focus on complex statistics, or concentrate on specific frameworks or vendors.

We ARE

Instead, we are focused on teaching AI decision-making, promoting systems thinking, and helping you explain AI concepts clearly and effectively to others.

Defining AI

Artificial Intelligence systems excel at learning from data, making predictions, and enhancing performance through accumulated experience.



Understanding AI Abilities



Learn Patterns

AI identifies trends and relationships in data.



Make Predictions

AI forecasts outcomes based on analyzed information.



Improve Performance

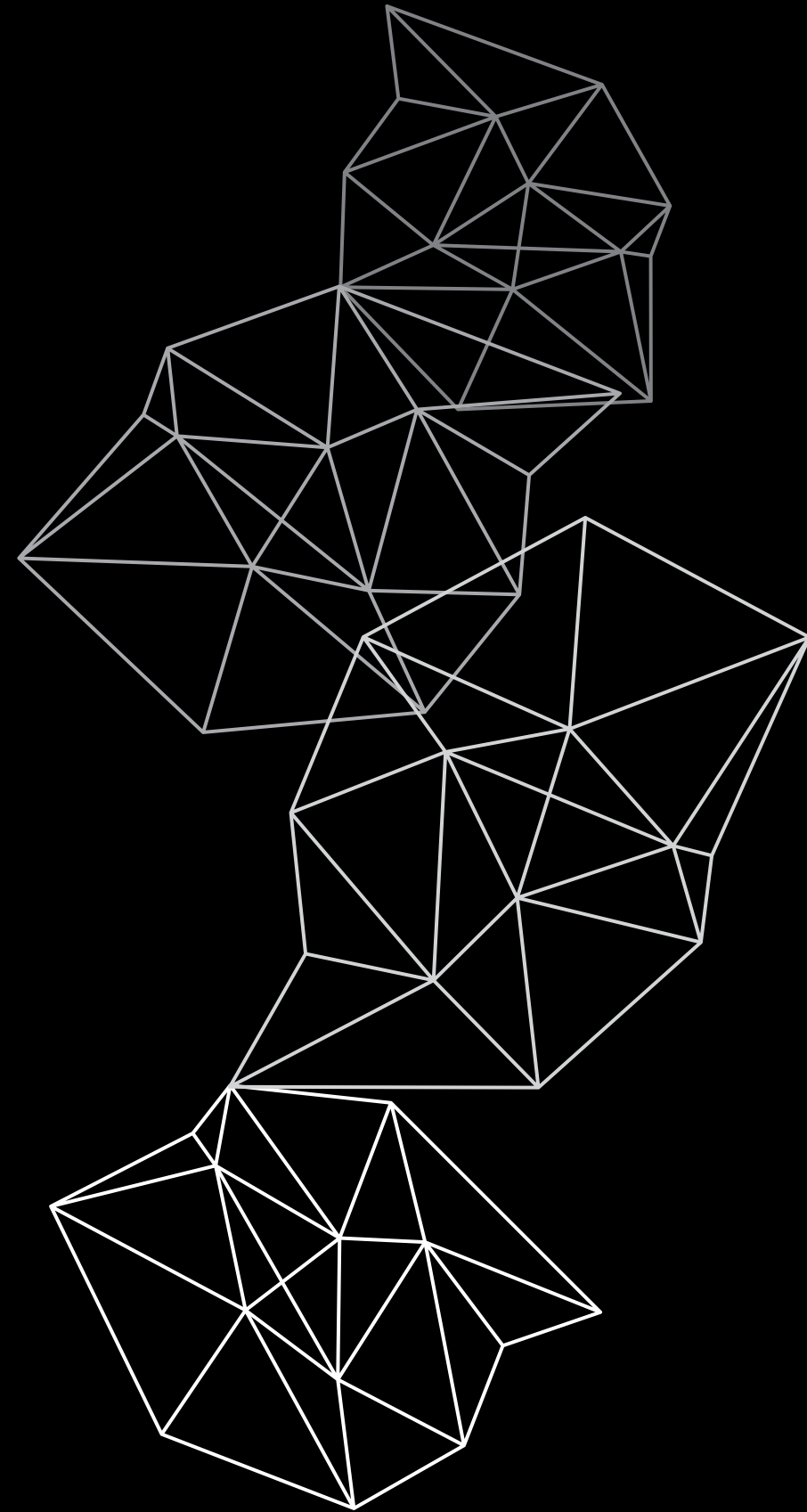
AI enhances processes through continual experience.



Optimize Systems

AI fine-tunes operations for better results.

Understanding AI Intelligence



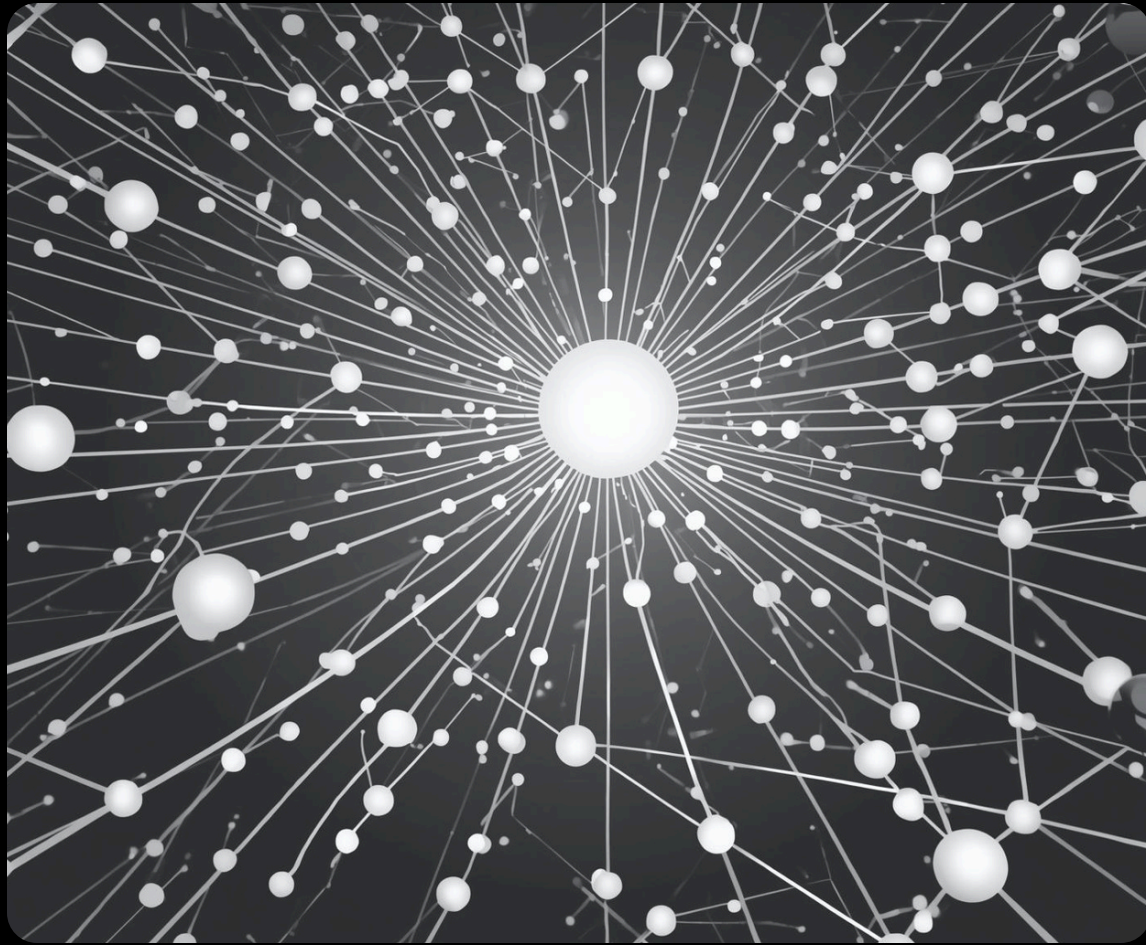
AI is NOT

AI lacks consciousness, self-awareness, and does not reason like humans; it operates on different principles.

AI IS

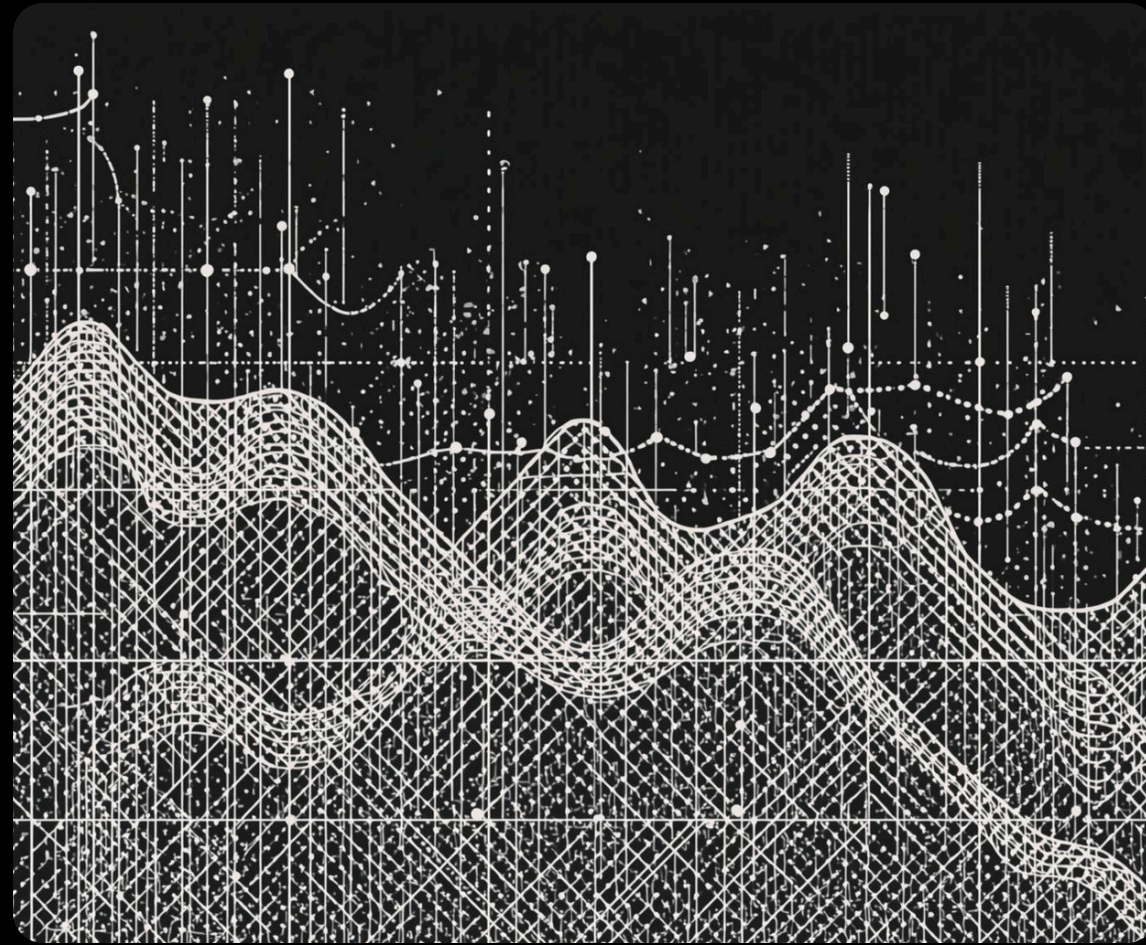
AI excels in large-scale statistical pattern recognition and optimization across vast datasets, enhancing efficiency significantly.

Understanding AI, ML, and Deep Learning



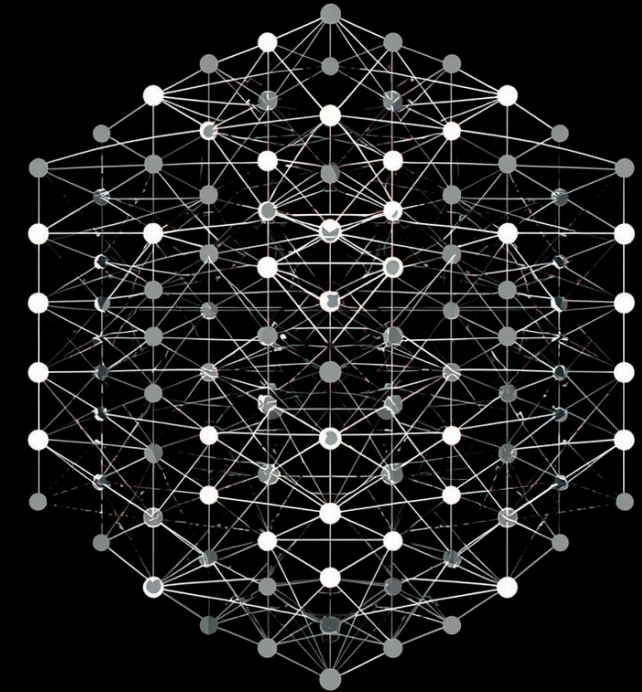
ARTIFICIAL INTELLIGENCE

Broad goal encompassing various technologies and methods.



MACHINE LEARNING

Learning from data to improve predictions and decisions.



DEEP LEARNING

Neural networks with many layers for deeper analysis.

Deep
Learning

LLMs are a type of
model

Trained on massive
text datasets

Specialized for
language tasks

Limitations

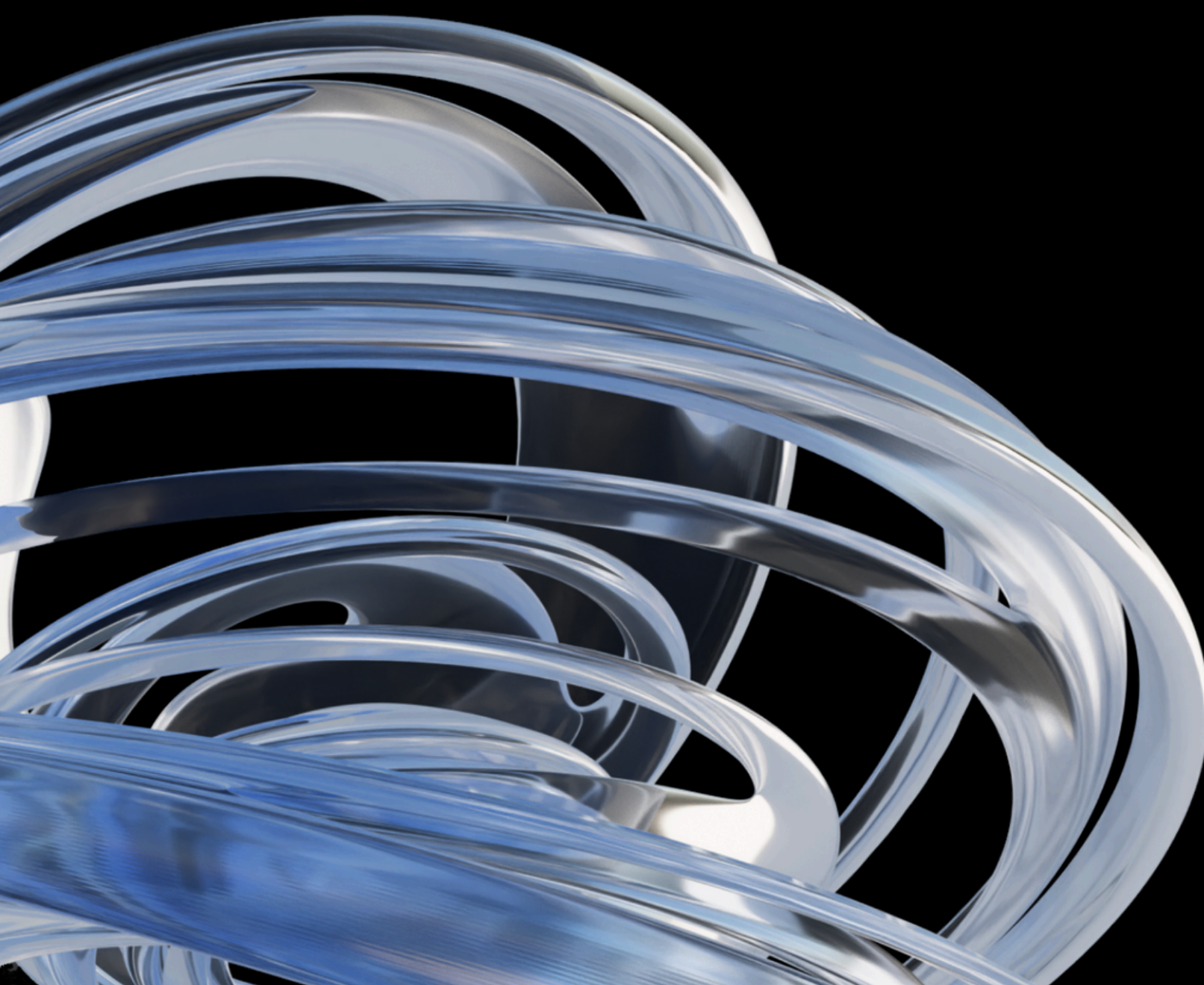
Limited to training
data scope

Cannot learn new
information easily

Understanding LLMs

This slide outlines where LLMs fit into the AI landscape, highlighting their capabilities and inherent limitations through a structured format of main points and supporting details.

LLMs Feel Intelligent



SPEAK FLUENTLY

Communicate clearly with natural language proficiency.

ADAPT ACROSS TASKS

Effectively transition between various tasks seamlessly.

MIMIC REASONING PATTERNS

Imitate logical processes without true comprehension.

What AI Is Bad At

POOR-QUALITY DATA

AI struggles significantly when faced with **no or low-quality data**, leading to inaccurate predictions and unreliable outcomes. Quality inputs are essential for effective AI performance and decision-making.

CHANGING LOGIC

Rapidly evolving logic presents a challenge for AI systems, as they often rely on fixed rules. This **instability** can result in errors or failures in output and decision processes.

AI Limitations

RARE EDGE CASES

AI often **struggles** to manage rare edge cases where data patterns are insufficient, leading to inaccurate predictions and increased error rates in decision-making processes.

TRUE REASONING

AI lacks the ability for **true reasoning and judgment**, which are essential for tasks requiring deep understanding, context, and critical thinking beyond its training data.

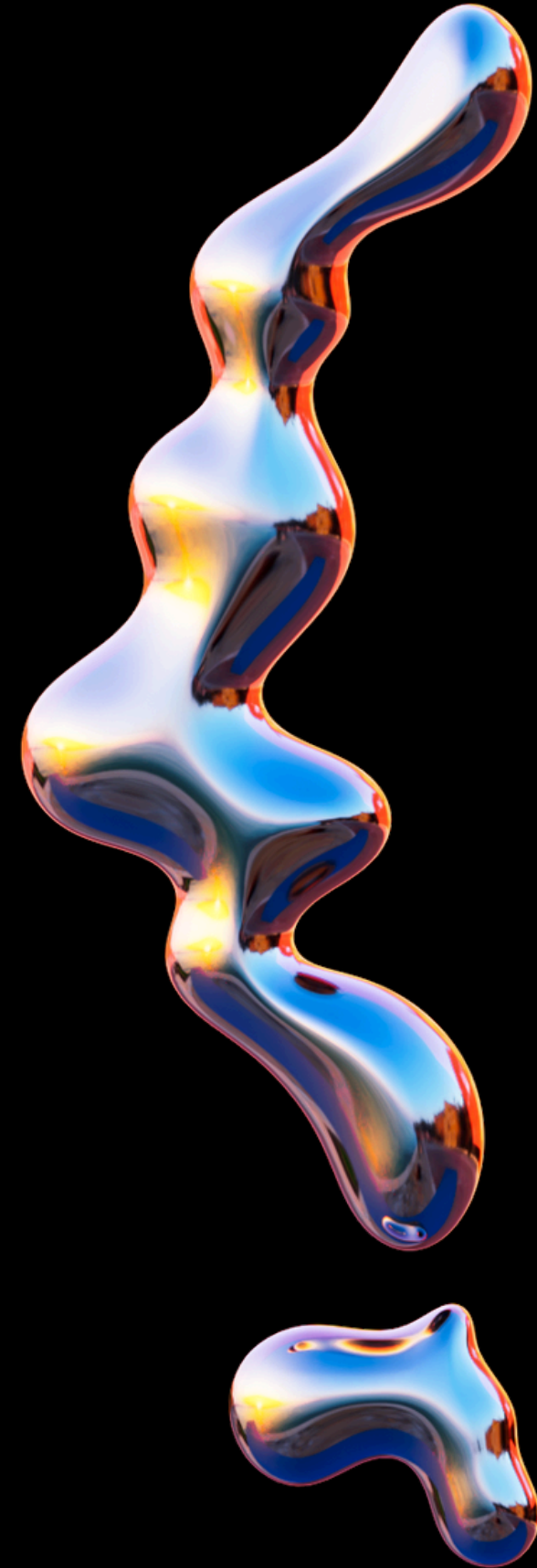
Example: Spam Detection

This section highlights how AI effectively categorizes email content into **spam or not spam**, showcasing the **classification** capabilities of AI without the need for deep learning.



Example: Price Prediction

Price prediction involves using historical data and market signals to generate **numeric values** for future prices. This method showcases AI capabilities without relying on deep learning techniques.



AI Systems vs Traditional Software

Traditional Software

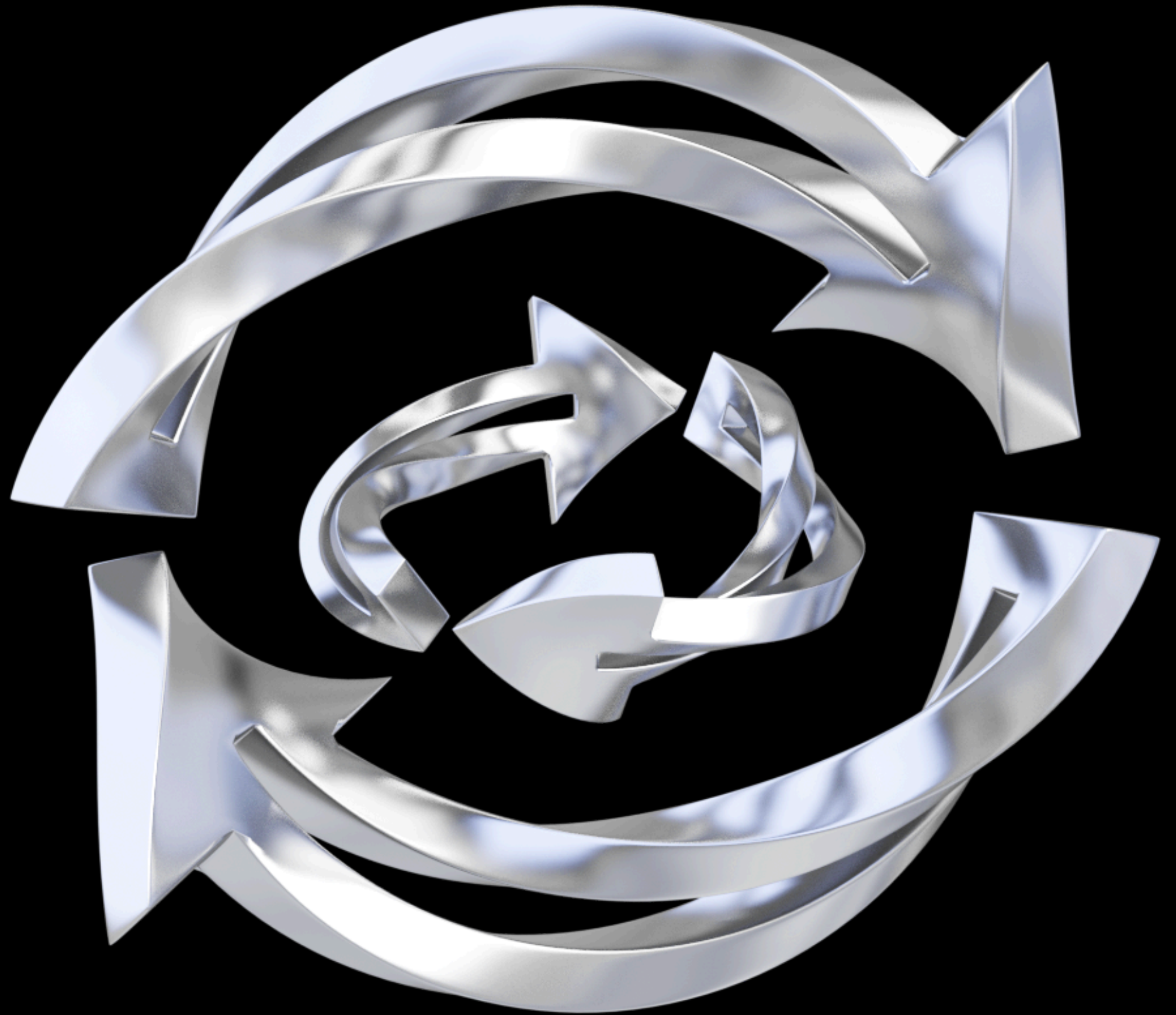
Rule-based systems that are deterministic with predictable failures, making them easier to understand and manage.

AI Systems

Data-driven and probabilistic systems that lead to uncertain outcomes, presenting unique challenges in engineering and deployment.

The Mental Shift

Embracing AI engineering means understanding that it involves **decision-making** rather than mere coding. Selecting the right problem and approach is crucial; a wrong choice can render any model ineffective, emphasizing the need for strategic thought in AI development.



Group Exercise

Analyzing AI Problem Scenarios

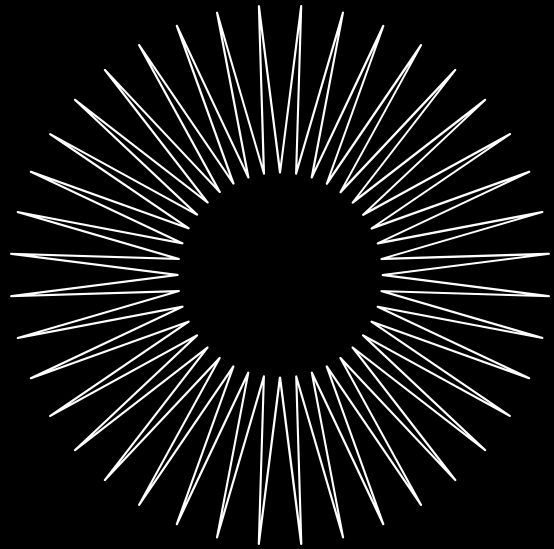
Is this AI?

For password validation, the problem is based on fixed rules, making it a traditional software challenge rather than an AI problem.

Why or Why Not?

Fraud detection uses data patterns, showcasing its nature as an AI problem that requires understanding complex behaviors and probabilities.

Key Takeaways



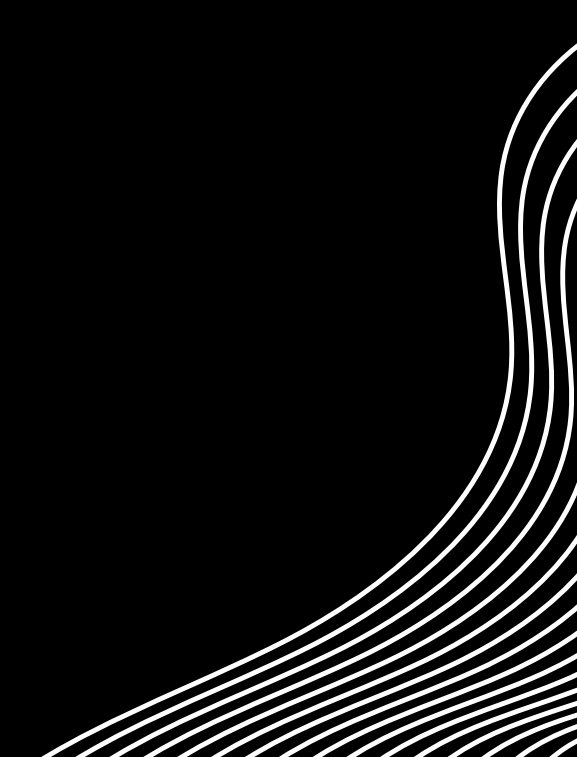
Important Points about AI Systems

1. AI learns patterns from data
 2. LLMs are a subset of AI
 3. Most AI failures happen before coding
 4. **Clarity beats cleverness**
-

What Comes Next

Upcoming Sessions on Machine Learning Topics

The next sessions will delve into **how machine learning actually learns**, explore classification versus regression, and provide insights on **correctly identifying AI problems** in your work.



Reflection / Homework

Prepare for the next session

Before the next session, please watch the video: **StatQuest – What is Machine Learning?** and think about where **AI should NOT be used** in your current work.

